Vectors in n Dimensions

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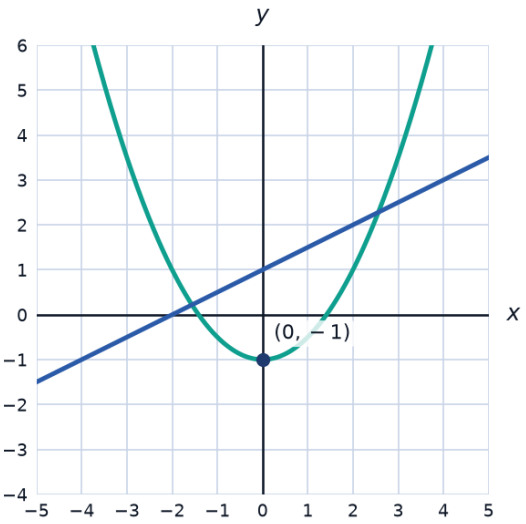
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider two vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

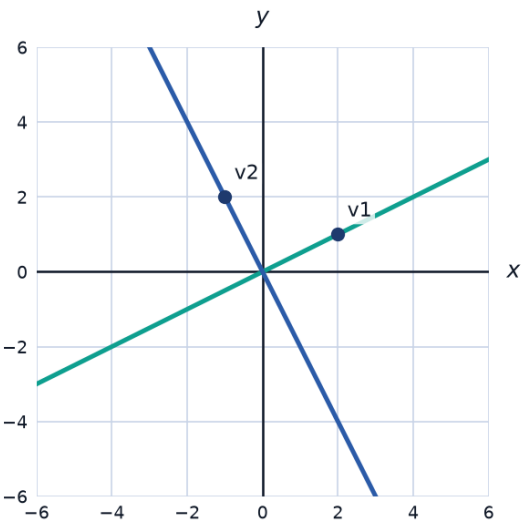
Consider the vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Express the complete solution set of the non-homogeneous linear system in parametric vector form:

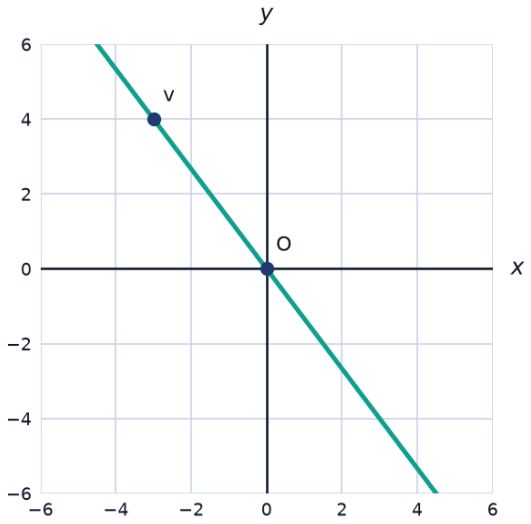
$\{x_1 + 2x_2 - x_3 + 3x_4 = 4; 2x_1 + 4x_2 - x_3 + 8x_4 = 11; -x_1 - 2x_2 + 2x_3 - x_4 = -1$



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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector $\mathbf{v} = (-3; 4)$ in $\mathbb{R}^2$. What is the magnitude (or norm) of this vector?

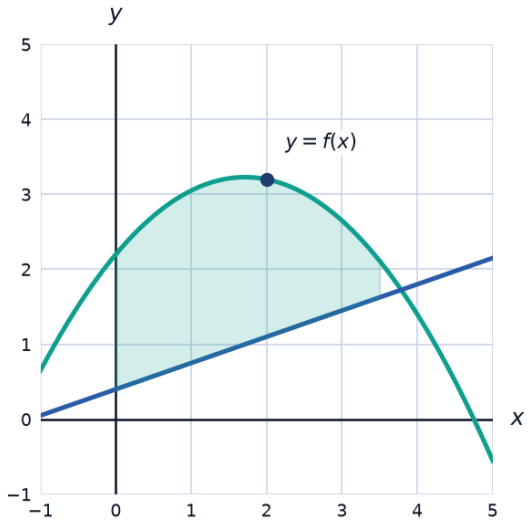


5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector $\mathbf{a} = (2\ k)$ and the vector $\mathbf{b} = (-1\ 3)$. \\ Part 1: If $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, find the value of k. \\ Part 2: Let the resulting vector from Part 1 be $\mathbf{v}$. The transformation T is defined by the matrix $M = (1, 2\ 2, -1)$. \\ If $T(\mathbf{v}) = (x\ y)$, find the value of x.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

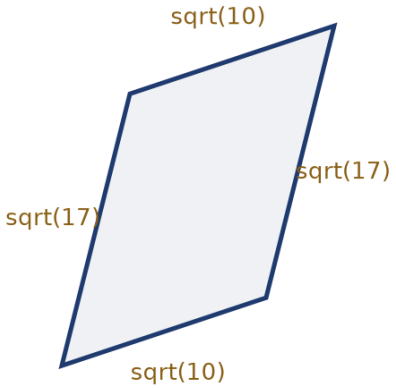
A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.



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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

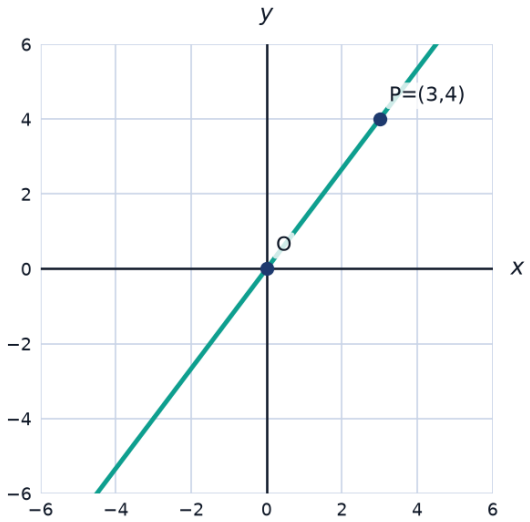
A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the vector $\mathbf{v} = (3; 4)$. If this vector is plotted on a standard *xy*-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?



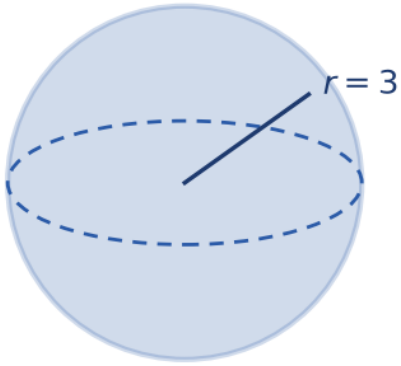
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the parallelepiped in $\mathbb{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$, and $\mathbf{w} = (3, 4, 1)$. The volume of this parallelepiped is given as 15 cubic units. Find the possible value(s) of k.

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a parallelepiped in $\mathbb{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3,\ -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ \ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the initial value problem for the coupled system of linear differential equations:

$\mathbf{x}'(t) = (1, 4; 2, 3)\mathbf{x}(t), \qquad \mathbf{x}(0) = (5; 1).$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. Furthermore, let W be the subspace of $\mathbb{R}^2$ defined by $W = \{(a, b) \in \mathbb{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Prove the Parallelogram Law for arbitrary vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $||\mathbf{u} + \mathbf{v}||^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2||\mathbf{u}||^2 + 2||\mathbf{v}||^2$. What is its geometric significance?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

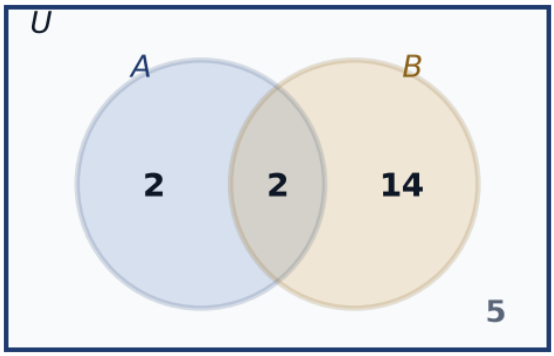
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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Determinants

The vector $\mathbf{v} = (-2; 1)$ is an eigenvector of the matrix $A = (3, 2; 1, 4)$. What is the corresponding eigenvalue λ?

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Determine, with formal verification via the Subspace Test, whether $W = \{A \in M_2(\mathbb{R}) : A^T = -A\}$ (the set of skew-symmetric matrices) is a subspace of $M_2(\mathbb{R})$. Find a basis and $\dim(W)$.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Express the invertible matrix $A = (1, 2; 3, 4)$ as a product of elementary matrices.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Determinants

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Let $\mathbf{u} = (1, -2, 2, 0)^T$ and $\mathbf{v} = (2, 1, 0, -2)^T$ in $\mathbb{R}^4$. Compute $||\mathbf{u}||$, $||\mathbf{v}||$, the inner product $\mathbf{u} \cdot \mathbf{v}$, the cosine of the angle θ between them, and verify the Cauchy–Schwarz inequality.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (3, 1; 1, 3)$. Identify one of the eigenvalues of A.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\mathbf{u} = [2, -1, 4]$ and $\mathbf{v} = [-3, 5, 2]$ in $\mathbb{R}^3$, compute $3\mathbf{u} - 2\mathbf{v}$.

a. $[12, -13, 8]$

b. $[0, 7, 16]$

c. $[12, -7, 8]$

d. $[-12, 13, -8]$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Is the vector $\mathbf{w} = [7, 4]$ a linear combination of $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, -1]$?

a. Yes, $\mathbf{w} = \frac{19}{7}\mathbf{v}_1 + \frac{10}{7}\mathbf{v}_2$

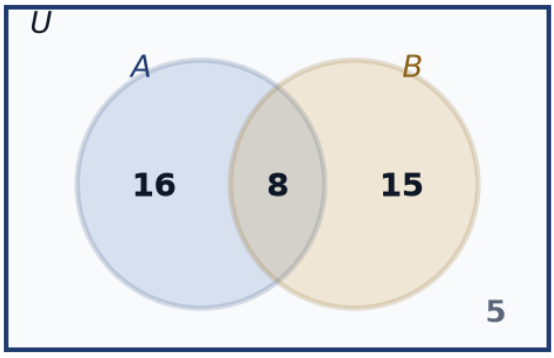
b. No, the vectors are linearly dependent

c. Yes, $\mathbf{w} = 2\mathbf{v}_1 + \mathbf{v}_2$

d. No, because the determinant is zero

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the definition of the span of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in $\mathbb{R}^n$?



a. The set of all possible linear combinations $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k$ where $c_i \in \mathbb{R}$

b. The set of all vectors perpendicular to $\mathbf{v}_1$

c. The sum of the lengths of the vectors

d. The matrix formed by using the vectors as columns

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if and only if:

a. $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$ implies $c_1 = c_2 = \ldots = c_k = 0$

b. At least one vector is a multiple of another

c. The dot product of any two vectors is zero

d. The number of vectors k is greater than the dimension n

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Can any set of 4 vectors in $\mathbb{R}^3$ be linearly independent?

a. No, any set of more than n vectors in $\mathbb{R}^n$ is guaranteed to be linearly dependent

b. Yes, provided none of the vectors is the zero vector

c. Yes, if they all have unit length

d. Only if their cross products are non-zero

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Are the vectors $\mathbf{v}_1 = [1, 2, 0]$, $\mathbf{v}_2 = [0, 1, 3]$, and $\mathbf{v}_3 = [2, 5, 3]$ linearly independent?

a. No, because $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$

b. Yes, they span all of $\mathbb{R}^3$

c. No, because they contain a zero entry

d. Yes, because their determinant is 1

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the Euclidean norm (length) of the vector $\mathbf{v} = [2, -3, 6]$ in $\mathbb{R}^3$.

a. 7

b. $\sqrt{11}$

c. 49

d. $\sqrt{41}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the unit vector in the direction of $\mathbf{u} = [-1, 2, 2]$.

a. $\left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

b. $[-1, 2, 2]$

c. $\left[-\frac{1}{9}, \frac{2}{9}, \frac{2}{9}\right]$

d. $\left[-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right]$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What does the Cauchy-Schwarz inequality state for any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$?

a. $|\mathbf{u} \cdot \mathbf{v}| \leq ||\mathbf{u}||||\mathbf{v}||$

b. $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$

c. $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||$

d. $|\mathbf{u} \cdot \mathbf{v}| \geq ||\mathbf{u}||||\mathbf{v}||$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the formula for the orthogonal projection of vector $\mathbf{y}$ onto a non-zero vector $\mathbf{u}$?

a. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}$

b. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{||\mathbf{y}||}\mathbf{u}$

c. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = (\mathbf{y} \cdot \mathbf{u})\mathbf{u}$

d. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \mathbf{y} - \mathbf{u}$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Compute the projection of $\mathbf{y} = [7, 6]$ onto $\mathbf{u} = [4, 2]$.

a. $[8, 4]$

b. $[4, 2]$

c. $[7, 6]$

d. $[2, 1]$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}}(\mathbf{y})$, what is the geometric relationship between $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ and $\mathbf{u}$?

a. $\mathbf{z}$ is orthogonal to $\mathbf{u}$

b. $\mathbf{z}$ is parallel to $\mathbf{u}$

c. $\mathbf{z}$ has unit length

d. $\mathbf{z} = \mathbf{0}$ always

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. Which of the following statements about $\mathbf{u}$ and $\mathbf{v}$ is true?

a. $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

b. The angle between $\mathbf{u}$ and $\mathbf{v}$ is acute.

c. The angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.

d. $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the matrix $A = (1, 1; 1, -1)$. Let $\mathbf{v} = (3; 1)$. Suppose we want to find the vector $\mathbf{x}$ of the form $c_1\mathbf{a}_1 + c_2\mathbf{a}_2$ where $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis derived from the columns of A, such that $||\mathbf{v} - \mathbf{x}||$ is minimized. Find the value of $||\mathbf{x}||$.

a. $\sqrt{10}$

b. $\sqrt{20}$

c. $\sqrt{5}$

d. $2\sqrt{10}$

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (1, 2, 3; 4, 5, 6; 7, 8, 9)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. V is a subspace of $\mathbb{R}^3$. Find the dimension of the orthogonal complement of V, denoted by $V^{\perp}$.

a. 1

b. 2

c. 0

d. 3

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a vector $\mathbf{v} = (3\ k)$ in $\mathbb{R}^2$. If the magnitude of $\mathbf{v}$ is $\sqrt{13}$, find the value of k given that $k > 0$.

a. 1

b. 2

c. 3

d. 4

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

a. 10

b. 2

c. -2

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Determinants

Consider the vectors $\mathbf{u} = (2, -3)$ and $\mathbf{v} = (4, 1)$ in $\mathbb{R}^2$. What is the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. 5

b. 8

c. 4

d. 11

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

Consider the matrix $A = (1, 2; -2, k)$. If the column vectors of A are orthogonal, find the value of k.

a. 2

b. -1

c. 4

d. -2

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Which of the following is a subspace?

a. x

b. $\{p \in P_2 : p(0) = 1\}$

c. x

d. $\{A \in M_{2\times 2} : \det A = 0\}$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Determinants

Consider the vector $\mathbf{v} = (5; -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as x.

a. 13

b. 17

c. 7

d. 169

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

Which of the following is NOT a subspace?

a. x

b. $\{A \in M_{2\times 2} : A^T = A\}$

c. x

d. x

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Let B be the basis $\langle -1, -1, 1\rangle, \langle -1, -2, 2\rangle, \langle -2, 1, -2\rangle$ of R^3. Find $[v]_B$ for $\mathbf{v} = \langle 8, 3, -1\rangle$.

a. $\langle 8, 3, -1\rangle$

b. $\langle 2, 1, 9\rangle$

c. $\langle -3, -1, -2\rangle$

d. $\langle 3, 1, 2\rangle$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Determinants

Consider a linear transformation represented by the matrix $A = (3, 0; 0, -2)$. If $\mathbf{v} = (1; 1)$ is an eigenvector of A, what is the corresponding eigenvalue?

a. 3

b. -2

c. 1

d. 5

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Systems of Linear Equations

Consider the matrix $A = (k, 2; 1, -3)$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

a. 4

b. 5

c. 3

d. 6

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